

BUSS386 Problem Set 3 — Solutions

Forwards and Futures

Prof. Ji-Woong Chung

Problem 1 — Classification

- (i) **Forward.** Exporter is *short* USD (sells USD, receives KRW). *OTC* — bilateral with the bank, customized notional and date.
- (ii) **Futures.** *Long* (buyer of the index). *Exchange* (KRX) by construction.
- (iii) **Futures.** Farmer is *short* (sells wheat). *Exchange* (CBOT/CME).
- (iv) **Forward.** Jeweller is *long* (buys gold). *OTC* — direct with a dealer, custom size.
- (v) **Futures.** *Short* S&P 500 e-mini to neutralize a long ETF position. *Exchange* (CME).

Problem 2 — Margin accounts and daily settlement

Multiplier $M = 250,000$. Daily P&L $= M \cdot (F_t - F_{t-1})$.

(a) Daily P&L:

- Day 1: $250,000 \times (408 - 405) = +750,000$.
- Day 2: $250,000 \times (402 - 408) = -1,500,000$.
- Day 3: $250,000 \times (393 - 402) = -2,250,000$.
- Day 4: $250,000 \times (397 - 393) = +1,000,000$.

(b) End-of-day balance (before any margin top-up):

Day	0	1	2	3	4
Balance (₩M)	10.00	10.75	9.25	7.00	(→ see (c))

(c) Margin call on **Day 3**: balance ₩7.00M is below maintenance ₩7.50M.

Required deposit to restore initial margin: $10,000,000 - 7,000,000 = 3,000,000$.

After the deposit, Day 3 balance returns to ₩10M. Day 4 close: $10 + 1 = 11$ M.

(d) Cash flows:

- Total cash deposited: 10M (initial) + 3M (call) = 13M.
- Cash returned at close (Day 4 balance): 11M.
- Overall P&L: $11 - 13 = -2,000,000$.

Cross-check: cumulative P&L $= 0.75 - 1.50 - 2.25 + 1.00 = -2.00$ M. ✓

Problem 3 — Long hedge with futures

- (a) Korean Air will *buy* fuel $\Rightarrow$ **long** the futures.
 Number of contracts = $200,000/25,000 = \boxed{8}$ contracts at \$90.
- (b) Total cost = (cost at spot) – (long futures gain) = $S_T \cdot Q - (S_T - F_0) \cdot Q = F_0 \cdot Q$.
 With $F_0 \cdot Q = 90 \times 200,000 = \boxed{\$18,000,000}$ for every $S_T \in \{\$84, \$90, \$96\}$.
 Quick check at $S_T = \$84$: $84(200,000) - (84 - 90)(200,000) = 16.8\text{M} + 1.2\text{M} = \18M . ✓
- (c) Algebraically the spot moves cancel: long-futures gain offsets the spot move dollar-for-dollar (perfect hedge assumption).
- (d) Realistic imperfections: (*any one of*) the May futures may not perfectly track the actual jet-fuel grade Korean Air burns; futures expiry may be a few days off; physical jet fuel may be priced at a regional differential to the futures benchmark; or the contract size doesn't divide evenly into the hedge requirement.

Problem 4 — Short hedge, and the regret that follows

- (a) The farmer will *sell* corn $\Rightarrow$ **short** the futures.
 Contracts = $250,000/5,000 = \boxed{50}$ short at \$4.60.
- (b) Revenue = (sale at spot) + (short futures gain) = $S_T Q + (F_0 - S_T)Q = F_0 Q$:

$$4.60 \times 250,000 = \boxed{\$1,150,000} \quad \text{in all three scenarios.}$$

Check at $S_T = \$5.10$: $5.10(250,000) - (5.10 - 4.60)(250,000) = 1,275,000 - 125,000 = \1.15M .
 ✓

- (c) The unhedged neighbour receives $S_T \times 250,000$:

S_T	\$4.20	\$4.60	\$5.10
Farmer (hedged)	\$1.150M	\$1.150M	\$1.150M
Neighbour (unhedged)	\$1.050M	\$1.150M	\$1.275M
Neighbour – farmer	–\$0.40/bu	\$0	+\$0.50/bu

The neighbour is better off **only at \$5.10**, by $\boxed{\$0.50/\text{bushel}}$ — and worse off by \$0.40/bu at \$4.20.

- (d) No. The hedge did exactly what it was bought to do: it converted an uncertain revenue into a certain \$1.15M, fixed in July when the farmer still had to commit to the crop. The \$0.50/bushel is not a loss but the *realised price of that certainty*, and it is only visible with hindsight — had corn fallen to \$4.20, the same hedge would have looked brilliant. Judging a hedge by the single path that happened to occur is precisely the error behind Cathay Pacific's 2020 fuel losses (Lec 1).

Problem 5 — Basis risk on USD/KRW

- (a) Samsung will receive USD and convert to KRW $\Rightarrow$ **short** USD/KRW futures (lock the KRW amount per USD).
 Contracts = $10,000,000 / 50,000 = \boxed{200}$ short.
- (b) On 31 May:
- Futures P&L (short, gains when futures falls): $(F_{\text{Mar}} - F_{\text{May}}) \cdot \$10\text{M} = (1,390 - 1,370) \cdot 10,000,000 = 200,000,000$ gain.
 - Spot conversion: $1,340 \cdot 10,000,000 = 13,400,000,000$.
 - Total KRW: $13,400,000,000 + 200,000,000 = 13,600,000,000$.
 - Effective rate per USD: $13,600,000,000 / 10,000,000 = \boxed{1,360}$ per USD.
 Equivalently: $S_{\text{May}} + (F_{\text{Mar}} - F_{\text{May}}) = 1,340 + 20 = 1,360$. ✓
- (c) On 30 June (futures expiry), $F_T = S_T$ exactly (basis = 0). Effective rate = $S_T + (F_{\text{Mar}} - S_T) = F_{\text{Mar}} = \boxed{1,390}$. The hedge perfectly locks the March futures price.
- (d) **Basis** = $S_t - F_t$, the gap between spot and futures. On 31 May the basis was $1,340 - 1,370 = -30$, while at expiry it would have been zero. Rearranging the effective rate shows exactly where it bites:

$$\underbrace{S_{\text{May}} + (F_{\text{Mar}} - F_{\text{May}})}_{\text{effective rate}} = F_{\text{Mar}} + \underbrace{(S_{\text{May}} - F_{\text{May}})}_{\text{basis at close}} = 1,390 + (-30) = 1,360.$$

So the hedge locks in the March futures price *plus the basis on the day you close* — and that basis is unknown on 1 March. *Basis risk* is precisely that residual uncertainty, and it arises whenever the hedge horizon does not coincide with futures expiry. Hold to expiry and the basis is zero, which is case (c).

Problem 6 — Cross hedge / minimum-variance hedge ratio

- (a) $h^* = \rho \cdot \frac{\sigma_S}{\sigma_F} = 0.85 \cdot \frac{0.025}{0.030} = 0.85 \cdot 0.8333 = \boxed{0.7083}$.
- (b) The airline is *short* 4,000,000 gallons of jet fuel (it still has to buy them), so it hedges with a **long** heating-oil futures position, which gains if fuel prices rise.

$$\text{Contracts} = \frac{h^* \cdot 4,000,000}{42,000} = \frac{2,833,333}{42,000} \approx 67.46 \Rightarrow \boxed{67 \text{ long contracts}}.$$

- (c) Hedge effectiveness = $R^2 = \rho^2 = (0.85)^2 = \boxed{72.25\%}$.
 Equivalently, 27.75% of the original variance remains, even with the optimal ratio.
- (d) Heating-oil price changes do not move 1:1 with jet-fuel price changes ($\rho < 1$). The component of jet-fuel price variation that is uncorrelated with heating oil cannot be hedged by any heating-oil position — this is residual basis risk.
- (e) The minimum-variance target is $h^* \times 4,000,000 = 2,833,333$ gallons of coverage, but 67 contracts provide

$$67 \times 42,000 = 2,814,000 \text{ gallons,}$$

leaving $2,833,333 - 2,814,000 = \boxed{19,333 \text{ gallons}}$ uncovered — about 0.48% of the 4,000,000-gallon exposure. (Rounding *up* to 68 contracts would overshoot the target by 22,667 gallons, so 67 is the nearer choice.)

This is not basis risk. Basis risk is the imperfect co-movement of heating oil and jet fuel on the gallons that *were* hedged, and it is what ρ governs. The 19,333 gallons instead carry outright jet-fuel price risk, because nothing at all was traded against them. The two effects add, and no choice of h^* removes the second one — only a contract size that divided the exposure evenly would.

Problem 7 — Hedging and risk measures

Throughout, revenue is measured *per barrel* and the 5% worst outcome is $\mu + \sigma \Phi^{-1}(0.05) = \mu - 1.645\sigma$.

- (a) **Unhedged.** Revenue $= S_T \sim N(92, 8^2)$, so $\mu = \$92$, $\sigma = \$8$, and

$$5\% \text{ worst} = 92 - 1.645(8) = \boxed{\$78.84}.$$

- (b) **Perfect hedge.** Revenue $= S_T + (F_0 - S_T) = F_0 = \92 , a constant.

$\mu = \$92$, $\sigma = \boxed{\$0}$, and the 5% worst outcome is also $\boxed{\$92}$ — there is no distribution left.

- (c) **Cross hedge.**

(i) $h^* = \rho \frac{\sigma_S}{\sigma_C} = 0.75 \cdot \frac{8}{10} = \boxed{0.60}.$

- (ii) Revenue $= S_T - h^*(C_T - 78)$. Since $\mathbb{E}[C_T] = 78$, the futures leg is worth zero in expectation and $\mu = \boxed{\$92}$. For the variance,

$$\sigma^2 = \sigma_S^2 + h^{*2}\sigma_C^2 - 2h^*\rho\sigma_S\sigma_C = 64 + 0.36(100) - 2(0.6)(0.75)(8)(10) = 64 + 36 - 72 = 28,$$

so $\sigma = \sqrt{28} = \boxed{\$5.29}.$

Equivalently $\sigma = \sigma_S\sqrt{1 - \rho^2} = 8\sqrt{0.4375} = \$5.29. \checkmark$

- (iii) $5\% \text{ worst} = 92 - 1.645(5.29) = \boxed{\$83.30}.$

Hedge effectiveness $= \rho^2 = \boxed{56.25\%}$ of variance removed.

- (d) Side by side:

	Unhedged	Cross hedge	Perfect hedge
Mean	\$92.00	\$92.00	\$92.00
St. dev	\$8.00	\$5.29	\$0.00
5% worst case	\$78.84	\$83.30	\$92.00

Hedging does not raise the refiner's *expected* revenue — all three means are \$92. What it buys is the removal of downside: the 5% worst case improves by \$4.46/bbl with the crude hedge and by \$13.16/bbl with the perfect one. Using crude instead of gasoline forfeits about half of that, because only $\rho^2 = 56\%$ of the gasoline price variance is explained by crude; the remaining 44% is basis risk that no crude position can touch.

Problem 8 — Speculation and leverage

- (a) Contract notional = $1,000 \times 250,000 = \boxed{250,000,000}$.
 Contracts needed = $1,000,000,000/250,000,000 = \boxed{4}$ long contracts.
- (b) Initial margin = $10\% \times 1,000,000,000 = \boxed{100,000,000}$.
- (c) Index +3% is +30 index points. Futures P&L = $30 \times 250,000 \times 4 = 30,000,000$; the ETF gains $3\% \times 1,000\text{M} = 30,000,000$. *The same money* — the exposure is identical.

	Cash committed	Profit	Return on cash
(A) ETF	¥1,000,000,000	+¥30,000,000	$\boxed{+3\%}$
(B) Futures	¥100,000,000	+¥30,000,000	$\boxed{+30\%}$

- (d) Index −3% is −30 points, so the P&L is −30,000,000 in both cases:

	Cash committed	Loss	Return on cash
(A) ETF	¥1,000,000,000	−¥30,000,000	$\boxed{-3\%}$
(B) Futures	¥100,000,000	−¥30,000,000	$\boxed{-30\%}$

The leverage is symmetric: it multiplies the loss by exactly the same factor as the gain.

- (e) Margin is exhausted when cumulative losses reach ¥100,000,000:

$$\Delta \times 250,000 \times 4 = 100,000,000 \Rightarrow \Delta = \boxed{100 \text{ index points}},$$

i.e. the index falling from 1,000 to 900, a drop of $\boxed{10\%}$. Note this is just $1/\text{leverage} = 1/10$. In practice she is in trouble well before then: the *maintenance* margin triggers a call after a far smaller move, and she must fund it the same day.

- (f) The extra return is not skill and not a different view — it comes entirely from committing ¥100M instead of ¥1,000M for the same ¥1,000M of exposure. That is leverage of $1,000/100 = 10\times$, and it scales gains and losses alike. What she gives up is staying power: a 10% adverse move erases her whole stake, and daily margin calls can force her out of the position *before* the index turns her way — exactly the mechanism that broke the UK LDI funds in Lec 2, on a smaller stage.